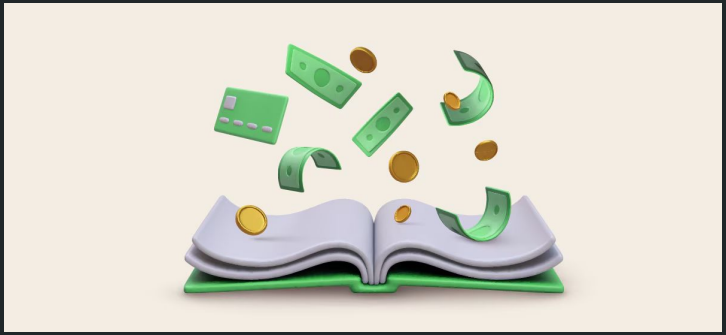
WealthMate



Aleksa Mihajlovic

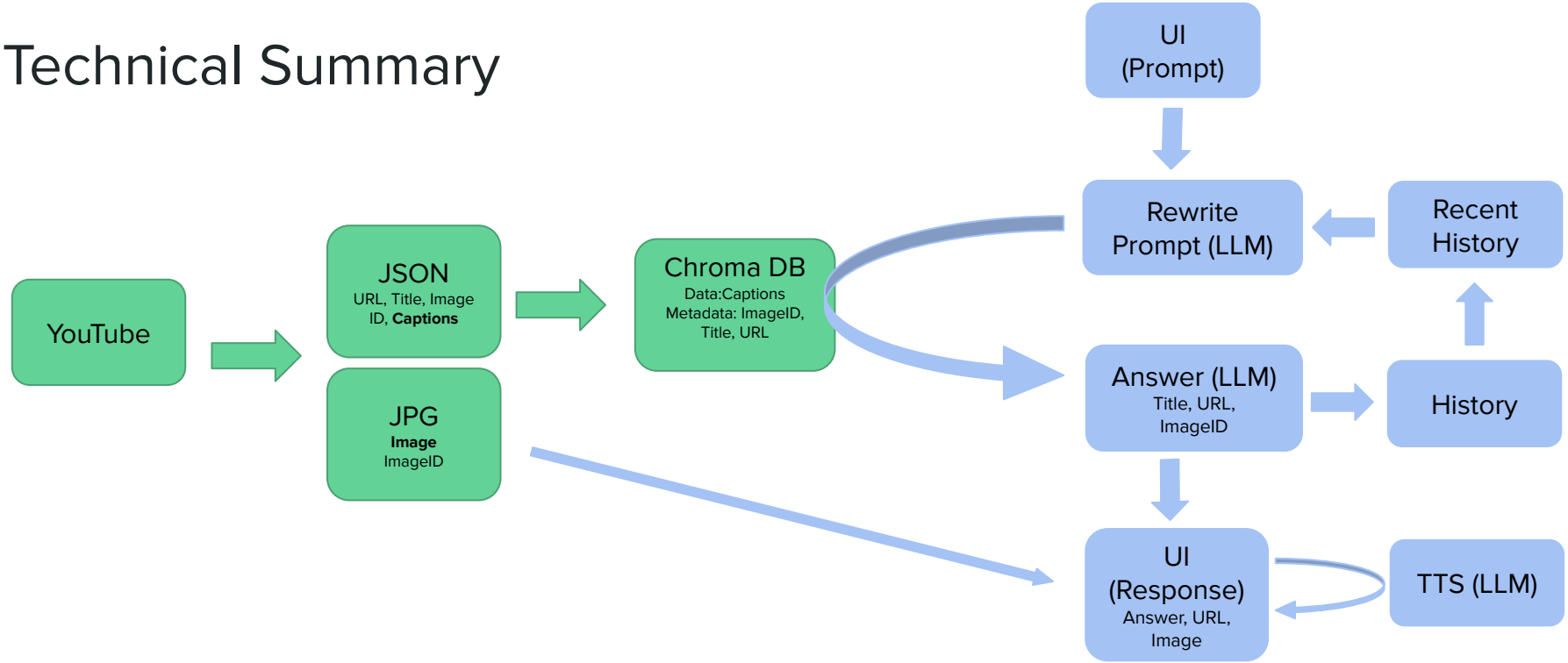
WealthMate

- Your friendly finance teacher that explains financial concepts in simple language
- Powered by trusted YouTube financial education sources
- Uses AI to deliver accurate, grounded, and easy-to-understand answers
- Designed to help people learn investing without jargon and confusion

Dataset

- Youtube channel **Plain Bagel** (100+ videos)
- Retrieval with **yt_dlp** package: Url, Title, VideoID, Captions, Thumbnails
- Preprocessing: None
- Data loaded as a JSON file and JPGs
- Stored in **Chroma DB** (Embedded: Captions; Metadata: URL, VideoID, Title)
- Chunk Size 800, overlap 150 (worked best)

Technical Summary



- Deployment: Remote Server/Web App

Q&A Agent

```python
rewrite_prompt = ChatPromptTemplate.from_messages([
    ("system", """
        Rewrite the user's latest question ONLY if needed.

        Rules:
        1. If the user's question already makes sense as an investing/finance question,
        return it EXACTLY as it is (no changes).
        2. If the user's question is about the assistant itself (name, identity, abilities),
        return it EXACTLY as it is (no changes).
        3. If it does NOT make sense as an investing question, rewrite it so it fits
        the context of the last 3 human messages.
        4. Never answer the question. Only rewrite it.
        5. If rewriting is impossible, return: "Unable to rewrite."

        Last 3 human messages: {history}
    """),
    ("human", "{input}")
])
rewrite_chain = rewrite_prompt | llm | RunnableLambda(lambda x: x.content.strip())
```

```python
answer_prompt = ChatPromptTemplate.from_messages([
    ("system", """
        You are WealthMate, a friendly, beginner-focused investing assistant.
        If the user asks about your name, identity,
        or abilities, answer normally without using retrieved documents.

        Otherwise, you answer ONLY using the retrieved documents.
        If the answer is not there, apologize and say you don't know the answer.
        If the answer is there, answer the question, but don't mention you are
        referring to the context. Finish answers politely by suggesting that the
        user watch the approved YouTube sources.
    """),
    ("human", "Question: {question}\n\nContext:\n{context}\n\nAnswer:")
])
```

```python
rag_chain = (

    RunnableLambda(lambda x: {
        "question_rewritten":
            rewrite_chain.invoke({"input": x["question"],
            "history": last_three_human(x["history"])})
    })

    | RunnableLambda(lambda x: {
        "question": x["question_rewritten"],
        "docs": retriever.invoke(x["question_rewritten"])
    })

    | RunnableLambda(lambda x: {
        "question": x["question"],
        "docs": x["docs"],
        "context": build_context(x["docs"])
    })

    | RunnableLambda(lambda x: {
        "question": x["question"],
        "docs": x["docs"],
        "answer":
            llm.invoke(answer_prompt.format(question=x["question"], context=x["context"])).content
    })
)

return rag_chain
```

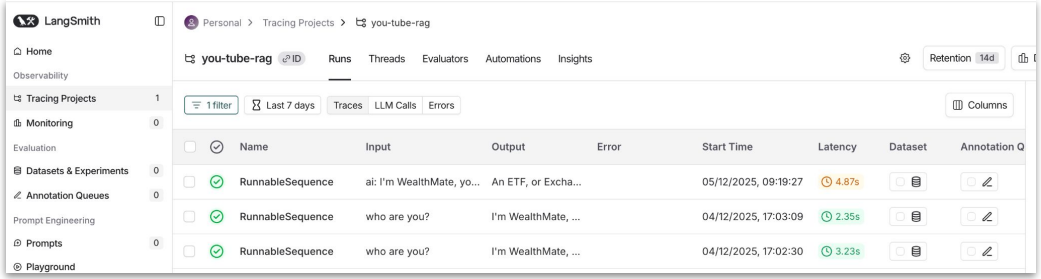
Q&A Agent - Audio Output

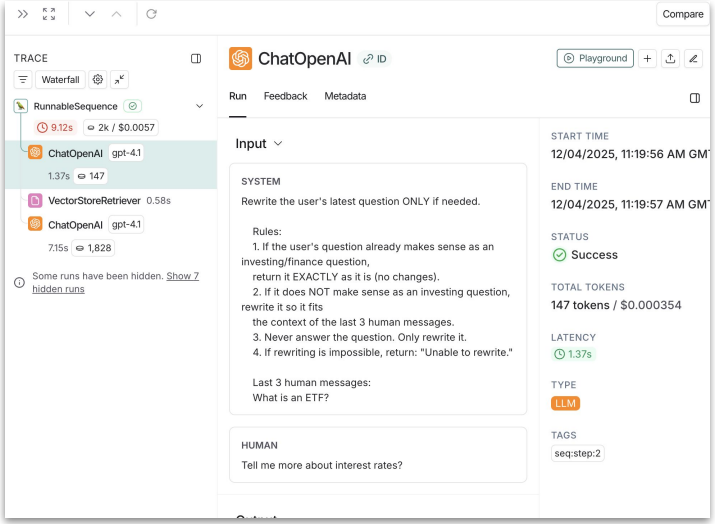
- Eleven Labs: (voice cloning, too expensive)
- Open Voice, Chatterbox (local models, voice cloning, compatibility issue)
- gpt-realtime (Websocket needed)
- gpt-4o-mini-tts (raw, too long to respond)
- **gpt-4o-mini-tts (Batching the answer text into multiple parts -> TTS)**

Tools

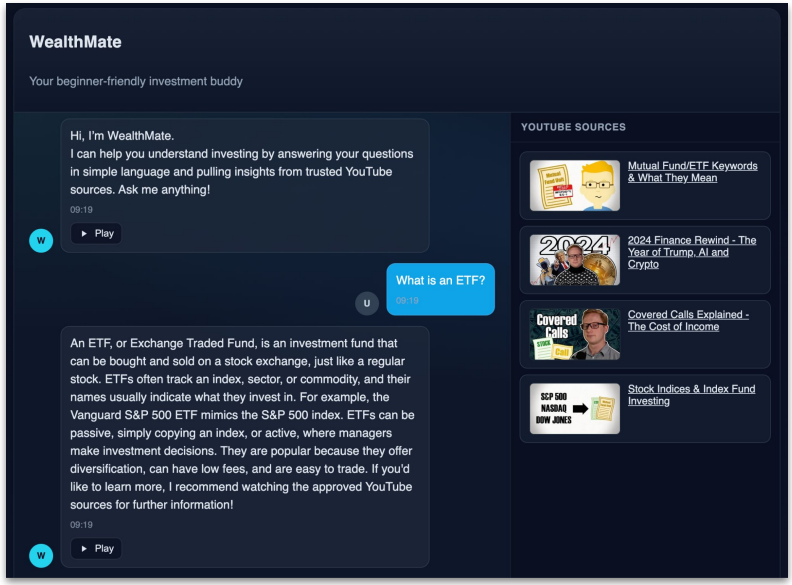
- Python 3.12 + FastAPI + Next.js
- Chroma DB Vector database
- LLMs: gpt-4o-mini, gpt-4o-mini-tts
- Langchain
- Langsmith

Evaluation


LangSmith
Personal > Tracing Projects > you-tube-rag
Home
Observability
Tracing Projects 1
Monitoring 0
Evaluation
Datasets & Experiments 0
Annotation Queues 0
Prompt Engineering
Prompts 0
Playground
you-tube-rag ID
Runs
Threads
Evaluators
Automations
Insights
Retention 14d
1 filter
Last 7 days
Traces
LLM Calls
Errors
Columns
Name
Input
Output
Error
Start Time
Latency
Dataset
Annotation Q
RunnableSequence
ai: I'm WealthMate, yo...
An ETF, or Excha...
05/12/2025, 09:19:27
4.87s
RunnableSequence
who are you?
I'm WealthMate, ...
04/12/2025, 17:03:09
2.35s
RunnableSequence
who are you?
I'm WealthMate, ...
04/12/2025, 17:02:30
3.23s


Compare
TRACE
Waterfall
RunnableSequence
9.12s
2k / $0.0057
ChatOpenAI gpt-4.1
1.37s 147
VectorStoreRetriever 0.58s
ChatOpenAI gpt-4.1
7.15s 1,828
Some runs have been hidden. Show 7 hidden runs
ChatOpenAI ID
Playground
Run
Feedback
Metadata
Input
SYSTEM
Rewrite the user's latest question ONLY if needed.
Rules:
1. If the user's question already makes sense as an investing/finance question,
return it EXACTLY as it is (no changes).
2. If it does NOT make sense as an investing question, rewrite it so it fits
the context of the last 3 human messages.
3. Never answer the question. Only rewrite it.
4. If rewriting is impossible, return: "Unable to rewrite."
Last 3 human messages:
What is an ETF?
HUMAN
Tell me more about interest rates?
START TIME
12/04/2025, 11:19:56 AM GMT
END TIME
12/04/2025, 11:19:57 AM GMT
STATUS
Success
TOTAL TOKENS
147 tokens / $0.000354
LATENCY
1.37s
TYPE
LLM
TAGS
seq:step:2

Conclusion & Demo

- Reliable and functional RAG App
- Interactive UI and Audio support
- Improvement: More functionality (investment advices)

https://maleckicoa.com/demo-apps/you-tube-rag